INFINITESIMAL LIFTING CRITERION FOR ALGEBRAIC STACKS

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2034-2043 of sources/stacks-stacks-more-morphisms/pinned-original-source.tex; proof span is lines 2045-2095, the complete author proof (49 body lines).

Editorial note. Navigation stubs: 10 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 19 of queue_stacks-stacks-more-morphisms.json (pb=2268, stmt 2034-2043, proof 2045-2095).

Editorial note. stacks-stacks-more-morphisms wave smm01 (2026-10-14): fresh adjudication accept per lane-s3/smm_judgments.md; pool per Chief 10:5x open-pool ruling (net 59 with-proof rows, bloodline lane-s3 by anchor audit, v2 true-key namespace stacks:stacks-more-morphisms.tex, row lemma-deform-property-fp-over-ft avoided on cross-chapter key collision); 60 with-proof rows judged 45 accept / 14 reject / 1 avoid; patch-sd-envs contract spans scanned.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 045F: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2095–2095. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06D2: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2095–2095. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06D3: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2095–2095. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma equation-diagram: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1896–1903. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Date: stacks-project:stacks-more-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma lemma-lift-to-smooth: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1991–1998. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma lemma-reformulate-formal-smoothness: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 1937–1942. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 04AM: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2095–2095. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 075W: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2095–2095. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 06Q3: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2095–2095. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0DNP: sources/stacks-stacks-more-morphisms/pinned-original-source.tex, source lines 2095–2095. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1 (Infinitesimal lifting criterion)**.** *Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks. The following are equivalent:*

(1) *The morphism f is smooth.*
(2) *The morphism f is locally of finite presentation and formally smooth.*

Proof. Assume f is smooth. Then f is locally of finite presentation by Morphisms of Stacks, Lemma 0DNP. Hence it suffices given a diagram (equation-diagram) and a $\gamma : y \circ i \to f \circ x$ to find a dotted arrow (see Lemma lemma-reformulate-formal-smoothness). Forming fibre products we obtain

$$\begin{array}{ccccc} T & \longrightarrow & T' \times_{\mathcal{Y}} \mathcal{X} & \longrightarrow & \mathcal{X} \\ \downarrow & & \downarrow & & \downarrow \\ T' & \longrightarrow & T' & \longrightarrow & \mathcal{Y} \end{array}$$

Thus we see it is sufficient to find a dotted arrow in the left square. Since $T' \times_{\mathcal{Y}} \mathcal{X} \to T'$ is smooth (Morphisms of Stacks, Lemma 075W) existence of a dotted arrow in the left square is guaranteed by Lemma lemma-lift-to-smooth.

Conversely, suppose that f is locally of finite presentation and formally smooth. Choose a scheme U and a surjective smooth morphism $U \to \mathcal{X}$. Then $a : U \to \mathcal{X}$ and $b : U \to \mathcal{Y}$ are representable by algebraic spaces and locally of finite presentation (use Morphisms of Stacks, Lemma 06Q3 and the fact seen above that a smooth morphism is locally of finite presentation). We will apply the general principle of Algebraic Stacks, Lemma 045F with as input the equivalence of More on Morphisms of Spaces, Lemma 04AM and simultaneously use the translation of Criteria for Representability, Lemma 06D3. We first apply this to a to see that a is formally smooth on objects. Next, we use that f is formally smooth on objects

by assumption (see Lemma lemma-reformulate-formal-smoothness) and Criteria for Representability, Lemma 06D2 to see that $b = f \circ a$ is formally smooth on objects. Then we apply the principle once more to conclude that b is smooth. This means that f is smooth by the definition of smoothness for morphisms of algebraic stacks and the proof is complete. $\square$